PROJECT CASE STUDY

Voice-ID & Neural Voice Cloning

Speaker Authentication + Voice Synthesis

Category	Status	Year
Speech AI	Completed prototype	2026

An integrated audio system that authenticates enrolled speakers with ECAPA embeddings and generates cloned speech through a standardized multi-format pipeline.

0.4814	16 kHz	3+
calibrated threshold	audio standard	input formats

Context and problem

Voice applications need to distinguish enrolled users from unknown speakers while accepting inconsistent real-world audio formats.

Strategy and implementation

- Generated SpeechBrain ECAPA speaker embeddings.
- Calibrated cosine-similarity decisions rather than using an arbitrary threshold.
- Standardized OGG, MP3, and WAV audio to 16 kHz.
- Connected the identification stage with Coqui TTS voice cloning.

My role

Designed the end-to-end audio preprocessing, speaker matching, threshold calibration, and synthesis workflow.

Technology stack

Python | SpeechBrain | ECAPA-TDNN | Coqui TTS | FFmpeg | NumPy

Results and evidence

- Established a custom decision threshold of 0.4814 for enrolled-vs-unknown queries.
- Built a reusable multi-stage audio pipeline.
- Combined authentication and generation in one coherent system.

Key takeaways

- Calibration is as important as the embedding model.
- Audio normalization prevents many failures before inference begins.

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